

Table: IoT vs AI vs Data Analytics Comparison

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 05 of 12

Table: IoT vs AI vs Data Analytics Comparison

LEARN IT

These three technologies are often used together, so students mix them up. The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

WORKED EXAMPLE

One problem, three technologies — a K-Electric smart meter:

IoT: The smart electricity meter measures units used every 15 minutes and sends them over the mobile network. Output: raw readings.

Data Analytics: The company cleans and studies a month of readings and draws a graph showing that most electricity is used between 7 pm and 11 pm. Output: report/graph.

AI: Software learns each area's pattern, predicts tomorrow's demand, and decides how to plan load management. Output: smart decision.

REMEMBER IT

Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

COMMON MISTAKE

Mistake: Writing that IoT "makes decisions". **Fix:** In the comparison table IoT's main purpose is *data collection and automation*; the *intelligent decision* belongs to AI.

اردو میں سمجھیں

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔ AI ذہین سافٹ ویئر ہے جو فیصلے اور پیش گوئیاں کرتا ہے۔ ڈیٹا اینالٹکس ڈیٹا کا جائزہ لے کر رپورٹس اور گراف بناتا ہے۔ تینوں مل کر ایک اسمارٹ نظام بناتے ہیں۔

CHECK YOURSELF

1. Which of the three works mainly with hardware?
2. What is the output of Data Analytics?

Answers: 1. IoT. 2. Reports, graphs and insights.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Data analytics is:

- a. Collect, clean, study, present data for decisions
- b. Only drawing NAND
- c. Only cabling
- d. Only Figma

MCQ 2. Cleaning data means:

- a. Fixing missing/wrong values
- b. Washing the PC
- c. Defrag only
- d. Sorting OSI layers

MCQ 3. A dashboard is:

- a. A visual summary of analytics
- b. A primary key
- c. A sprint log only
- d. A K-map

MCQ 4. Analytics without a question is:

- a. Just a pile of charts
- b. Always AI
- c. Always IoT
- d. Always SQL keys

MCQ 5. Finding that "second shift fails more in CS" is:

- a. Insight from analysis
- b. A foreign key
- c. A gate delay
- d. A beachhead

MCQ 6. Python's later tool for this in XII is:

- a. Pandas + charts
- b. LogiSim
- c. Figma only
- d. Access macros only

Extra short questions

1. Define data analytics and list its steps.
2. Why clean first?
3. Give a college decision driven by analytics.
4. How is analytics different from IoT?

Extra long question

1. A principal has 3 years of BIEK results. Outline an analytics project: source, clean, one chart, one decision.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Collect → organise/clean → analyse → present → decide.
2. Garbage in, garbage out — mean of marks with blanks is wrong.
3. Extra classes for a subject whose fail rate jumped.
4. IoT gathers live device data; analytics is the studying/presenting (often of that data).

Long

1. Source: gazette CSVs. Clean: missing roll numbers. Chart: line of pass% by year.
Decision: change lab hours if practical marks drag theory. Mention ethics (no public shaming of named students).

Next lecture: 5.2 Uses of IoT in Different Areas